

Social Network Aggregation Beyond Preservation

From property preservation to structural repair

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Main idea of the talk

1. Preservation

- **Static** social network aggregation: can F preserve a property δ ?
- **Dynamic** aggregation: can preservation remain coherent under new information?

2. Aggregation as repair

- inputs may be **structurally deficient**
- aggregation as a **repair mechanism**: output better than each limited input?

Key inspiration: a Condorcet-like pattern is *paradoxical* for preferences, but **constructive** for social networks.

Background

Social network aggregation

Fix a set of actors A . Each individual reports a social network over the same actors:

$$E_1, \dots, E_n \subseteq A \times A.$$

A social network aggregation rule returns one collective network:

$$F : (2^{A \times A})^n \rightarrow 2^{A \times A}, \quad F(\mathbf{E}) \subseteq A \times A.$$

Why aggregate?

Individual networks may capture different family, workplace, online, or local-community relations. Aggregation builds a collective representation from these partial views.

Working interpretation: local social information $\rightsquigarrow$ collective structural judgement.

A methodology from social choice theory

- study aggregation rules **axiomatically**
- same style in *preference aggregation*, *graph aggregation*, and *argumentation-framework aggregation*
- **Unanimity**: all accept $e \Rightarrow$ accept e .
- **Groundedness**: none accept $e \Rightarrow$ reject e .
- **Independence**: each tie is decided only by its supporters.
- **Neutrality**: ties with the same support pattern are treated alike.

Question: axioms for F vs. properties of $F(\mathbf{E})$.

Social network properties as rationality conditions

In social network aggregation, a **network property** δ plays the role of a rationality condition on collective structure.

Property	Social-network meaning
Connectivity / six degrees	every pair of actors can reach each other
Resilience	backup support; no single fragile channel
Triadic closure	friends of your friends are also your friends
Weak ties / margin	Victor–Martin style emphasis on ties linking a group to the outside

$$E_i \models \delta \text{ for all } i \quad \implies \quad F(\mathbf{E}) \models \delta ?$$

Why preserve? The collective network will be used as a social map: preserving δ protects the structural role that made the inputs useful.

Tensions between axioms, properties, and repair

Tensions between two desiderata

well-behaved rules vs. structurally good networks

1. Static preservation

- Can aggregation rules with **standard axioms** preserve desirable properties?

2. Dynamic rationality

- Rationality is often treated *statically*.
- Turn to **dynamic response**: how should collective networks react to new information?

3. Aggregation as repair

- Aggregation not only *preserves*; it may **repair**.
- Can $F(\mathbf{E})$ be better than the input networks?

Static aggregation results

Preservation asks: if every input network has P , must $F(\mathbf{E})$ have P ?

Why it matters: these properties are not just graph-theoretic. They encode **reachability**, **local cohesion**, and **bridges between groups**.

Possibility some properties can be preserved by natural rules.

Example: **nomination** $F(\mathbf{E}) = \bigcup_i E_i$ preserves **connectivity**.

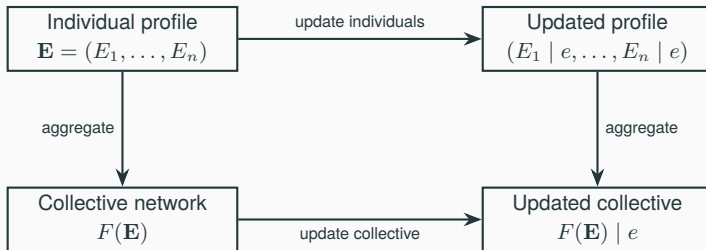
Intuition: union may add ties, but it does not delete paths.

Impossibility some structural properties clash with standard edge-wise axioms.

Example: preserving **triadic closure** may force oligarchic or dictatorial structure.

Intuition: edge-by-edge aggregation can mix fragments of different local patterns.

New information about a tie e : two routes should agree



$$F(E_1 | e, \dots, E_n | e) = F(E_1, \dots, E_n) | e.$$

Revision: accept a new tie

Contraction: reject an existing tie

Dynamic aggregation results

Dynamic preservation asks whether aggregation commutes with network change.

Why it matters: social networks are updated by new information. The collective network should not depend on whether we **update first** or **aggregate first**.

Possibility additive revision can behave well.

Example: **nomination** is dynamically rational for regular additive revision.

Intuition: adding a required tie commutes with taking the union.

Impossibility contraction and richer properties are harder.

Removing one tie can break a path, a triangle, or a bridge.

Intuition: avoiding order-dependence may require stable decisive agents.

Aggregation as Repair

From preservation to aggregation as repair

Reading	Formal pattern	Main question
Preservation	$\forall i, E_i \models \delta \Rightarrow F(\mathbf{E}) \models \delta$	Can aggregation keep a property already present in every input?
Repair	$\forall i, E_i \text{ deficient} \Rightarrow F(\mathbf{E}) \models \delta$	Can aggregation produce a better collective network than the inputs?

Aggregation as a structural repair mechanism.

What counts as a deficiency?

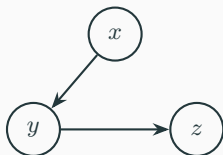
A **deficiency** is a structural problem in an individual social network.

Source	Examples	Repair reading
Failure of a property	fragmentation; excessive distance; open triads; missing bridges	output satisfies the target property δ although inputs fail it
Negative phenomenon	majority illusion : local view contradicts global structure	output removes a structural bias in the profile

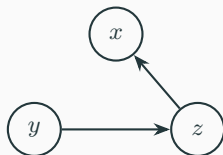
Example: failure of **triadic closure** in the Condorcet pattern.

Condorcet pattern: directed version

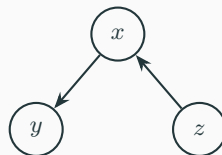
Majority over directed comparisons gives a cycle



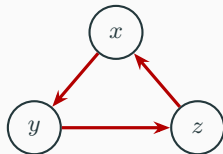
Agent 1: $x \succ y \succ z$



Agent 2: $y \succ z \succ x$



Agent 3: $z \succ x \succ y$

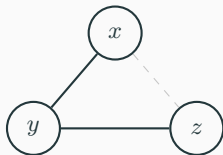


Majority cycle

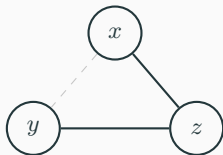
Preference reading: rational individual orders $\rightsquigarrow$ collective cycle.

Condorcet pattern: undirected version

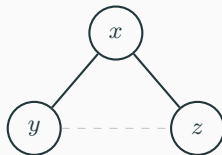
The same profile repairs triadic closure



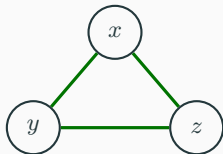
$$E_1 = \{xy, yz\}$$



$$E_2 = \{yz, zx\}$$



$$E_3 = \{zx, xy\}$$



Collective social network

Social-network reading: open triads $\rightsquigarrow$ completed triangle.

Repair result

Baseline result

- If all input networks are **the same graph**, aggregation returns that graph.
- If that graph has a deficiency, the collective network still has it.

$$E_1 = \dots = E_n = E \implies F(\mathbf{E}) = E \implies \text{no repair.}$$

Theorem

*No aggregation rule satisfying **unanimity** and **groundedness** can repair the network property of **independence** on all profiles.*

Boundary of non-repair

- Under unanimity, groundedness, independence, and neutrality, **dictatorship** is the unique rule with no repair capacity.
- This boundary is discussed for **majority illusion**; it is not a theorem for every kind of deficiency.

Two ways to make repair possible

Scope The following routes are mainly developed for **majority illusion**.

Route A: domain restriction

- restrict profiles to contain **complementary limitations**
- aim: allow better aggregation rules to repair
- example: **mixed profiles** for majority illusion

Route B: relax unanimity / groundedness

- strict boundaries may preserve local blind spots
- repair may require adding latent ties or dropping misleading ties

Kemeny-style approach

- choose a close collective network that is repair-safe
- fixed-test design reduces the complexity of a global Kemeny search

- **Static aggregation:** preservation of social-network properties; possibility and impossibility.
- **Dynamic aggregation:** rational reaction to new information; revision and contraction.
- **Aggregation as repair:** collective networks can be better than structurally limited inputs.

Further extension

- semantics under social networks; extending argumentation to positive/negative social relations.

About me

Current status

- Master's student at Sun Yat-sen University.
- Expected graduation: **June 2027**.
- Currently seeking a **doctoral position**.

Publications

- Yutong Yi and Weiwei Chen. *External Semantics for Argumentation Frameworks*. Proceedings of the 11th International Conference on Computational Models of Argument, IOS Press, 2026.
- Yutong Yi and Weiwei Chen. *Social Network Aggregation*. Submitted to *Logic Journal of the IGPL*; currently under revision.

Awards

- Best Young Scholar Contribution, 2025. Awarded by the BLESS Conference Committee.
- National Scholarship, 2025. Awarded by the Office of the Academic Degrees Committee of the State Council, China.

Thank you!

Questions and comments are welcome.